

Math 1232 Summer 2026

Midterm Exam

- You will have 90 minutes for this test.
- You are not allowed to consult books or notes during the test, but you may use a one-page, one-sided cheat sheet you have made for yourself ahead of time.
- You may use a four-function calculator. You may leave answers unsimplified, except you should compute trigonometric functions as far as possible.
- The exam has 10 problems, and is 6 pages total.
- Each problem is worth ten points. The whole test is scored out of 100 points.
- Read the questions carefully and make sure to answer the actual question asked. Make sure to justify your answers—math is largely about clear communication and argument, so an unjustified answer is much like no answer at all. When in doubt, show more work and write complete sentences.
- If you need more paper to show work, I have extra at the front of the room.
- Good luck!

1	2	3	4	5
6	7	8	9	10

Total /100

Name: **Solutions**

Problem 1. Compute $\int \frac{1}{x(2 + \ln(x))} dx$.

Solution. We can take $u = 2 + \ln(x)$ so that $du = 1/x dx$. Then

$$\begin{aligned}\int \frac{1}{x(2 + \ln(x))} dx &= \int \frac{1}{u} du \\ &= \ln |u| + C = \ln |2 + \ln(x)| + C.\end{aligned}$$

Problem 2. Compute $\frac{d}{dx} e^{\arccos(x^2)}$.

Solution.

$$\frac{d}{dx} e^{\arccos(x^2)} = e^{\arccos(x^2)} \cdot \frac{-1}{\sqrt{1 - (x^2)^2}} \cdot 2x = e^{\arccos(x^2)} \cdot \frac{-2x}{\sqrt{1 - x^4}}.$$

Problem 3. Let $h(x) = x^5 + x$. Compute $(h^{-1})'(2)$.

Solution. By the Inverse Function Theorem, we know that

$$(h^{-1})'(2) = \frac{1}{h'(h^{-1}(2))}.$$

Guess and check shows that $h(1) = 2$, so $h^{-1}(2) = 1$. And we know that

$$h'(x) = 5x^4 + 1$$

and thus

$$h'(1) = 5 + 1 = 6$$

Therefore

$$(h^{-1})'(2) = \frac{1}{6}.$$

Problem 4. Compute $\lim_{x \rightarrow \infty} x^{\frac{3}{x+2}}$.

Solution. We are looking at an indeterminate form of type ∞^0 , so we take logarithms. Setting L to be our limit gives

$$\ln(L) = \lim_{x \rightarrow \infty} \frac{3}{x+2} \ln(x) = \lim_{x \rightarrow \infty} \frac{3 \ln(x)}{x+2},$$

which is of type ∞/∞ , meaning we can apply l'Hôpital's rule:

$$\ln(L) = \lim_{x \rightarrow \infty} \frac{\frac{3}{x}}{1} = \lim_{x \rightarrow \infty} \frac{3}{x} = 0.$$

Then $\ln(L) = 0$ means $L = e^0 = 1$.

Problem 5. Compute $\int x e^{3x} dx$.

Solution. We integrate by parts, with $u = x$ and $dv = e^{3x}$. This gives $du = 1$ and $v = \frac{1}{3}e^{3x}$. The integration by parts formula then gives

$$\int x e^{3x} = \frac{x}{3} e^{3x} - \frac{1}{3} \int e^{3x} dx = \frac{x}{3} e^{3x} - \frac{1}{9} e^{3x} + C.$$

Problem 6. Compute $\int \frac{1}{x^2\sqrt{4+x^2}} dx$.

Solution. We see a $\sqrt{4+x^2}$ in the integrand, so we try the trig substitution $x = 2 \tan(\theta)$. Then $dx = 2 \sec^2(\theta) d\theta$ and we get

$$\begin{aligned}\int \frac{1}{x^2\sqrt{4+x^2}} dx &= \int \frac{1}{4 \tan^2(\theta) \sqrt{4+4 \tan^2(\theta)}} 2 \sec^2(\theta) d\theta \\&= \frac{1}{4} \int \frac{2 \sec^2(\theta)}{\tan^2(\theta) \sqrt{4(1+\tan^2(\theta))}} d\theta \\&= \frac{1}{4} \int \frac{2 \sec^2(\theta)}{\tan^2(\theta) \sqrt{4 \sec^2(\theta)}} d\theta \\&= \frac{1}{4} \int \frac{2 \sec^2(\theta)}{\tan^2(\theta) \cdot 2 \sec(\theta)} d\theta = \frac{1}{4} \int \frac{\sec(\theta)}{\tan^2(\theta)} d\theta \\&= \frac{1}{4} \int \frac{\frac{1}{\cos(\theta)}}{\frac{\sin^2(\theta)}{\cos^2(\theta)}} d\theta = \frac{1}{4} \int \frac{\cos(\theta)}{\sin^2(\theta)} d\theta.\end{aligned}$$

At this point we can set $u = \sin(\theta)$ to get $du = \cos(\theta) d\theta$ and our integral becomes

$$\frac{1}{4} \int \frac{1}{u^2} du = \frac{1}{4} \cdot \frac{-1}{u} + C = -\frac{1}{4 \sin(\theta)} + C.$$

Now we rewrite in terms of x . Since $x = 2 \tan(\theta)$ was our substitution, we have that $x/2 = \tan(\theta)$ is opposite over adjacent, so we draw a right triangle with side opposite θ equal to x and side adjacent to θ equal to 2. We want to calculate $\sin(\theta)$, which is opposite over hypotenuse, so we use the Pythagorean Theorem to find that the hypotenuse is $\sqrt{4+x^2}$. Then $\sin(\theta) = x/\sqrt{4+x^2}$, so that

$$\int \frac{1}{x^2\sqrt{4+x^2}} dx = -\frac{1}{4 \sin(\theta)} = -\frac{\sqrt{4+x^2}}{4x} + C.$$

Problem 7. Compute $\int \frac{10}{(x+2)(x-3)} dx$.

Solution. We need to do a partial fractions decomposition. We have

$$\begin{aligned}\frac{10}{(x+2)(x-3)} &= \frac{A}{x+2} + \frac{B}{x-3} \\ 10 &= A(x-3) + B(x+2). \\ x=3 : \quad 10 &= B \cdot 5 \\ B &= 2 \\ x=-2 : \quad 10 &= A \cdot -5 \\ A &= -2\end{aligned}$$

Thus we compute

$$\begin{aligned}\int \frac{10}{(x+2)(x-3)} dx &= \int \frac{-2}{x+2} dx + \int \frac{2}{x-3} dx \\ &= -2 \ln |x+2| + 2 \ln |x-3| + C.\end{aligned}$$

Problem 8. Compute $\int \sin^3(x) \cos^6(x) dx$.

Solution. Given a product of sines and cosines, we're looking for one of the terms to have an odd power. In this case, that's $\sin^3(x)$, so we rewrite it as

$$\sin^3(x) = \sin^2(x) \sin(x) = (1 - \cos^2(x)) \sin(x)$$

to make the substitution $u = \cos(x)$ with $du = -\sin(x) dx$ work out. We have:

$$\begin{aligned}\int \sin^3(x) \cos^6(x) dx &= \int (1 - \cos^2(x)) \cos^6(x) \sin(x) dx \\ &= - \int (1 - u^2) u^6 du = \int u^8 - u^6 du \\ &= \frac{u^9}{9} - \frac{u^7}{7} + C = \frac{\cos^9(x)}{9} - \frac{\cos^7(x)}{7} + C.\end{aligned}$$

Problem 9. Compute $\int_1^\infty \frac{\ln(x)}{x} dx$. Does the integral converge or diverge?

Solution. This is type I improper since the domain is unbounded, so we replace the upper limit of integration by t , compute the integral, and let $t \rightarrow \infty$. To compute the integral, we do the substitution $u = \ln(x)$ to get $du = (1/x) dx$.

$$\begin{aligned}\int_1^\infty \frac{\ln(x)}{x} dx &= \lim_{t \rightarrow \infty} \int_1^t \frac{\ln(x)}{x} dx \\ &= \lim_{t \rightarrow \infty} \int_0^{\ln(t)} u du = \lim_{t \rightarrow \infty} \frac{u^2}{2} \Big|_0^{\ln(t)} \\ &= \lim_{t \rightarrow \infty} \frac{(\ln(t))^2}{2} - \frac{0^2}{2} = \infty,\end{aligned}$$

so the integral diverges.

Problem 10. Set up (but don't compute!) an integral for the area of a surface obtained by taking the curve $y = \ln(x^3 + 1)$ from $x = 0$ to $x = 10$ and rotating around the x -axis.

Solution. We have $y' = \frac{3x^2}{x^3+1}$ so we get

$$\begin{aligned}SA &= \int_0^{10} 2\pi y \sqrt{1 + y'^2} dx \\ &= \int_0^{10} 2\pi \ln(x^3 + 1) \sqrt{1 + \frac{9x^4}{(x^3 + 1)^2}} dx.\end{aligned}$$